

Frisco ISD Science and Engineering Fair Abstract and Certification

Project Vel: Development of a Coordinate-Blind, Decentralized Multi-Agent Swarm Protocol Utilizing Spatial Gradient Repulsion and Kinetic Vector Logic for Autonomous 3D Volume Exploration in GPS-Denied Disaster Topologies

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Autonomous search and rescue in GPS - denied, collapsed urban environments requires rapid spatial coverage without centralized network infrastructure or fixed positioning systems. Traditional swarm architectures depend on global positioning or heavy communication bandwidth, introducing single - point failures and high latency. This project introduces Project Vel, a coordinate - blind, fully decentralized multi - agent swarm protocol designed for rapid 3D volume dominance in complex disaster topologies. The protocol utilizes two novel algorithmic frameworks: Map - Aware Gradient Repulsion (MAGR) for dynamic spatial distribution, and Kinetic Vector Logic (KVL) for local obstacle avoidance and momentum retention. Agents operate on local relative sensing without absolute spatial coordinates. Physical validation was conducted using an ESP32 micro - rover testbed equipped with Time - of - Flight (ToF) infrared sensor arrays communicating via peer - to - peer ESP - NOW serialization. Large - scale performance was evaluated through Python - based 3D Shared Spatial Memory voxel grid simulations 40 x 40 x 40 using Monte Carlo sweeps up to N = 500 agents. Empirical trial data demonstrated that the Project Vel protocol achieved a 90% volume dominance rate 41% faster than baseline 2D / random - walk exploration models while reducing inter - agent collision rates by 63%. Peer - to - peer packet transmission latency maintained a mean of 1.1 ms, allowing real - time adaptivity during simulated hardware node failures. The system maintained robust volume mapping without performance degradation in signal - blindspot conditions. In conclusion, Project Vel proves that coordinate - blind, decentralized swarm logic provides a scalable, resilient solution for real - time 3D search operations in GPS - denied emergency environments, eliminating reliance on external positioning infrastructure.

Category

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1. As a part of this research project, the student directly handled, manipulated, or interacted with (check all that apply):

☐ human participants

☐ potentially hazardous biological agents

☐ vertebrate animals

☐ microorganisms

☐ rDNA

☐ tissue
2. This abstract describes only procedures performed by me/us, reflects my/our own independent research, and represents one year’s work only.

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3. I/We worked or used equipment in a regulated research institution or industrial setting.

☐ yes
 ☒ no
4. This project is a continuation of previous research.

☒ yes
 ☐ no
5. My display board includes non-published photographs/visual depictions of humans (other than myself)

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6. I/We hereby certify that the abstract and responses to the above statements are correct and properly reflect my/our own work.

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